

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven and innovative individual with a deep commitment to continuous learning, adaptability and proactive approach. Coming from a rigorous mathematical background, I aspire to develop solutions for the real-world, impactful problems.

SKILLS

Languages: English (C1 [IELTS 8.0](#) )
Programming: Python, Julia, Javascript, MATLAB, C/C++, R, SQL, Java
Tools: Linux, Git, GitHub Actions, Bash, PyTorch, OpenCV, Docker, Quarto, SSH, Kubernetes, L^AT_EX

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
GPA: 4.85/5.0
Supervisor: [Márton Vaitkus](#) 
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 (**rank 3/34**)
Supervisor: [Galtbayar Artbazar](#) 
Thesis: [Evaluating land degradation using image processing methods](#) 

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern & Lab Lead

- Developed [Python CLI tools](#)  for multispectral drone image processing to contribute to a research effort to evaluate the ecological land degradation on a massive scale. Implemented multi-threading algorithm to cut processing times for the images by 80% while testing all the necessary steps for accurate vegetation index calculation.
- Architected a unified, well-documented data folder structure for research data backups, directly improving data retrieval efficiency and creating written guidelines for future data integration of the drone imagery.
- Organized weekly land degradation research and mathematical seminars by preparing the discussion materials, reading the relevant materials and performed numerical experiments using Jupyter notebooks.
- Participated in the field survey data collection and multispectral measurements; gained first-hand insight into ecological data collection for a better design of the drone imagery analysis pipeline drawing from domain-specific insights from ecological and geography researchers.

PROJECTS

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced machine learning demonstrations from Bishop's *Deep Learning Foundations and Concepts* book using **Julia** programming language with innovative interactive, reactivite programming environment ([Pluto.jl](#) ). Deployed as interactive website using GitHub Actions, and customized it using **Javascript**.

[aiSim LiDAR point cloud sensor fusion for 3D Gaussian Splatting](#)

- Completed a project to train 3D Gaussian Splatting model from autonomous driving simulation data, by converting aiSim autonomous driving data into Nerfstudio format. Demonstrated the colorization the LiDAR point cloud measurements of [aiSim](#)  simulator using the 4-way surround camera images in **Python**.

LEADERSHIP & AWARDS

Stipendium Hungaricum Scholarship Sep, 2025
Presentation: [First NUM-Yonsei Student Workshop on Mathematics and Computing](#)  July, 2025